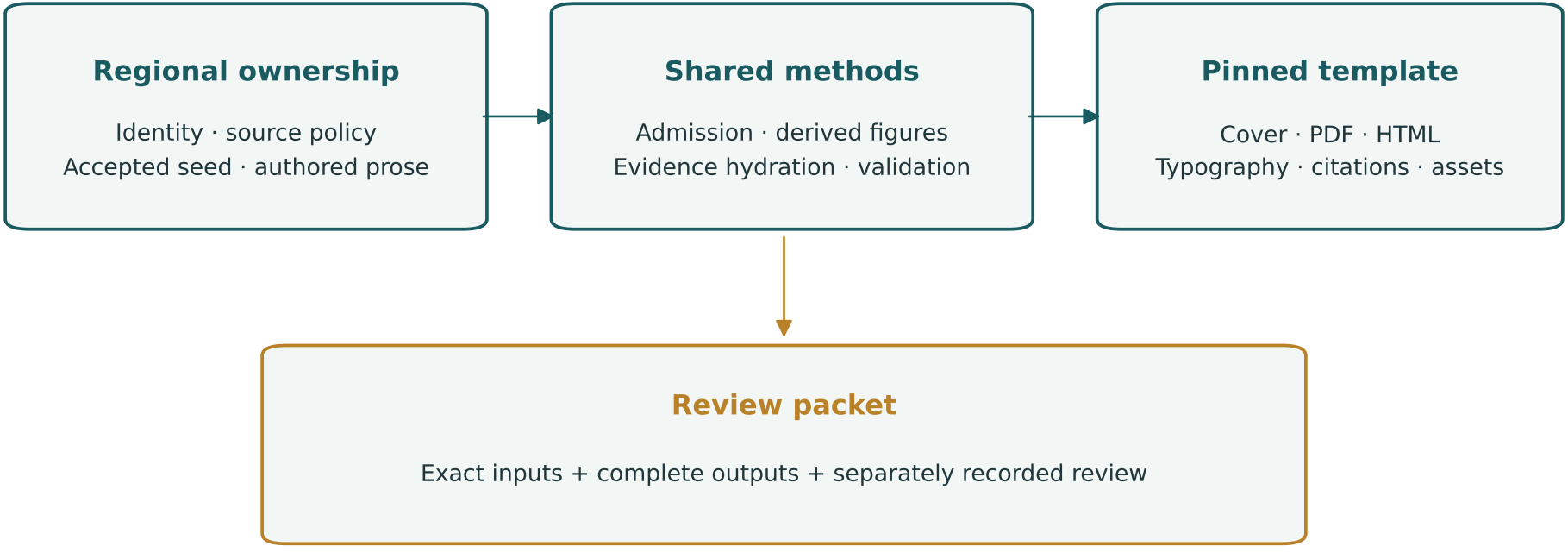Shared computation, explicit local evidence